

Research article

Sentiment analysis of Lovecraft's fiction writings

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ARTICLE INFO

Keywords:

Lovecraft
Plutchik's Wheel of emotions
Computational literary analysis
Sentiment analysis
Weird tales

ABSTRACT

This paper, at the intersection of literary studies, computational literary analysis and psychology, analyzes Lovecraft's literary production throughout his career. Fifty-four writings from the period 1905–1935, which span almost all of his career as a fiction writer, have been studied by applying the computational techniques of sentiment analysis and following a methodology structured in three stages. In the first one, I analyzed each of the writings, in order to identify the relative percentage of emotion-inducing words according to Plutchik's Wheel of Emotions. The second phase of this process consisted of hierarchical cluster analysis to identify homogeneous groups of works. Finally, in the third step of the research, I conducted a statistical analysis. The results point to three different conclusions. First, the quantitative analysis confirms that there exists an intimate connection between the emotions of fear and sadness in Lovecraft's writings, which generates an intense emotion of despair. Second, the vast majority of his works can be grouped into two categories, each with a distinct pattern in terms of emotions. Finally, there seems to be a relationship between the identified categories and the evolution of Lovecraft's style; that is, there is a temporal evolution toward much darker works, which are dominated by negative emotions.

1. Introduction

Lovecraft's work has performed as a bridge between low and high culture. On the one hand, his enormous impact on popular culture [1] has nurtured the inspiration of films, comics, music, and video games. On the other, his brand, derived from the communion of his work and his own personality, has aroused great interest in the academic community as well. As a result, we have witnessed the development of the most diverse research and even the existence of academic journals dedicated to the study of Lovecraft's works, such as *Lovecraft Studies* (1980–2005) and *Lovecraft Annual* 2007–present). However, his influence on popular culture is mainly due to the tales written in the last period of his literary production, when he created, sometimes in cooperation with other authors, a mythology of extraordinary creatures that has somehow shaped the general public's imagination.

There is no consensus as to how to classify Lovecraft's work, although most critics (see, for instance Refs. [3,4], suggest that his literary production can be grouped into three different periods: *the Macabre* (1905–1920), *the Dream Cycle* (1920–1927) and *the Cthulhu Mythos* (1925–1935). Having said that, it is “impossible to categorize the stories by their dates of creation, as they overlap, while containing elements that thematically fit a different cycle. *The Dream Cycle* and *the Cthulhu Mythos* share many aspects and themes, blurring the line between the categories further” ([5]; 48). Some tales written towards the end of his career are frequently included in the Dream Cycle, such as *At the Mountains of Madness* (1931), *The Dreams in the Witch House* (1933), or *The Thing on the Doorstep* (1933) (see the compilations of *Dream Cycle* stories selected by the publisher's Read & Co: Lovecraft and Weiss [6]).

Lovecraft's weird tales, which are the object of study in this paper, are remarkably difficult to define. Joshi [7] points out that “[...]”

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no definition of the weird tale embraces all types of works that can plausibly be assumed to enter into the scope of the term" (p. 2). Perhaps, one of the best approaches to the description of the weird tale is the one proposed by Lovecraft himself, who focused on what he calls *cosmic fear*: "The one test of the really weird is simply this - whether or not there be excited in the reader a profound sense of dread, and of contact with unknown spheres and powers" (Lovecraft [8][1927]; 7)

His weird stories "combine elements of the Gothic tradition with science fiction, fantasy and even some details that could be recognized in detective stories" ([9]; 111). They also present the insignificance of the human race as a sign of identity ([10]; 7). As Houellebecq notes about Lovecraft, "[f]ew beings have ever been so impregnated, pierced to the core, by the conviction of the absolute futility of human aspiration [...]. Everything will disappear. And human actions are as free and as stripped of meaning as the unfettered movement of the elementary particles" ([11]; 40–41). Certainly, as a preeminent Lovecraft authority, Joshi notes, "Lovecraft never passes up an opportunity to diminish human achievements" ([12]; no page), and the superiority of alien civilizations in diverse ways, from physiological to intellectual ([79]; 339). Despair in Lovecraft appears to be a consequence of the feeling of insignificance; as Rottensteiner ([13]; 121) points out "[...] the indifference of the cosmos and the insignificance of human beings in it, for HPL [Howard Phillips Lovecraft] a source of despair and imbued with an overwhelming negative feeling." Certainly, we may agree that "Lovecraft's characters are left in a state of utter despair as they try to grapple with the complete destruction of what was once their concept of reality" [15]; 161), an emotion that is transmitted to the reader. It will also be safe to assume that fear is clearly the dominant emotion in Lovecraft's work, given the very nature of the weird tale: "the oldest and strongest emotion of mankind is fear, and the oldest and strongest kind of fear is fear of the unknown. These facts few psychologists will dispute, and their admitted truth must establish for all time the genuineness and dignity of the weirdly horrible tale as a literary form" ([81], 1). With this particular vision of the weird tale, Lovecraft "created his own subgenre of horror –'cosmic horror' with its peculiarities, specific characters and mood. [...] The writer tried to create the specific and terrifying atmosphere both to scare and involve the recipient in further reading" ([16]; 47).

As I have mentioned already, the definition of what is a weird tale is a rather challenging task. However, Lovecraft's own description is based on the emotions the tale is supposed to produce, with a special emphasis on a specific type of fear, associated to our insignificant position in the cosmos:

"The true weird tale has something more than secret murder, bloody bones, or a sheeted form clanking chains according to rule. A certain atmosphere of breathless and unexplainable dread of outer, unknown forces must be present; and there must be a hint, expressed with a seriousness and portentousness becoming its subject, of that most terrible conception of the human brain—a malign and particular suspension or defeat of those fixed laws of Nature which are our only safeguard against the assaults of chaos and the daemons of unplumbed space." (Lovecraft [81]: 6)

Certainly, for Lovecraft, conveying some kind of emotion is essential, and as he points out, "[t]herefore we must judge a weird tale not by the author's intent, or by the mere mechanics of the plot; but by the emotional level which it attains at its least mundane point" ([81]; 7). To conduct such an analysis, it seems useful to frame this study of feelings in the conceptual framework of psychology of emotions, which provides a great in-depth assessment of them. I, therefore, follow the approach of [17]; 24–25), who concluded that "classification based on emotion words performs on par with traditional genre feature sets that are based on rich, open-vocabulary lexical information." In line with the method of [17]; I have chosen Plutchik's Wheel of Emotions [18]. In simple terms, this model establishes the existence of eight basic emotions: anger, anticipation, disgust, fear, joy, sadness, surprise, and trust. These primary emotions are organized in pairs of opposites: joy/sadness, fear/anger, anticipation/surprise, and disgust/trust. This approach also states that basic emotions can intensify, become weaker, or be combined to produce different emotional states. In fact, more complex emotions arise from a combination of two primary ones in dyads, so that, for example, the blending of joy and anticipation generates optimism. Similarly, this framework also contemplates the definition of more complex emotions arising from the union of three basic ones.

Two additional reasons justify this choice: this theory is the most commonly used in fine-grained sentiment analysis ([19]; 38), and the theoretical basis for the NRC Emotion Lexicon - EmoLex [20,21]) that I have applied in this paper. Since this lexicon has been used in hundreds of investigations, we can be confident of its reliability (for its application to literary texts see Refs. [22,23, and 24]. Other well-known approaches, such as Ekman's [25] six emotions model, have been more rarely employed in computational linguistics.

2. Emotions in Lovecraft's works: a literature review

The analysis, both direct or indirect, of the emotional dimension of Lovecraft's writing, has been addressed by several previous works with a qualitative approach, offering however quite a different angle than this paper attempts to do. Nevertheless, their conclusions are relevant indeed for this research, since they will allow to evaluate to what extent the results of this work are consistent with previous studies. Moreover, a greater benefit can be expected, for it will bring the opportunity to confront the concordance of qualitative and quantitative research. With this aim, the literature review that follows has gathered, on the one hand, the main contributions of the emotions that inhabit Lovecraft's tales, while on the other it has identified the stylistic devices used by the author to generate them.

2.1. The emotion of fear

The dominant emotion in Lovecraft's work is fear, as well as other emotions (dyads) that are associated with it according to Plutchik's theory, such as submission and despair. This fear is not physical since the subgenre of cosmic horror "exposes our deepest fears—not only fears of injury or death, but fears engendered by our ignorance of the fundamental nature of the world and the universe [...]" ([82]; x). The problem consists precisely of how to "depict with mere words that which forever lies beyond the sphere of the

known" ([26]; 189).

Certain elements are systematically used by Lovecraft to generate this peculiar feeling of fear. Andronova and Torhovets ([16]; 49) highlight that Lovecraft used epithets "in order to transfer the dark, mysterious and sinister atmosphere of his unique literary world by appealing to the reader's feeling of disgust, repulsion and fear," by using sight, smell, and hearing.

"I beheld the horned waning moon, red and sinister, quivering through the vapours that hovered over the distant valley of Banof." (Lovecraft [27],[1918]; 695)

"They said the place had a febrile smell." (Lovecraft [28],[1924]; 609)

"O friend and companion of night, thou who rejoicest in the baying of dogs (here a hideous howl burst forth) and spilt blood (here nameless sounds vied with morbid shriekings), who wanderest in the midst of shades among the tombs (here a whistling sigh occurred), who longest for blood and bringest terror to mortals (short, sharp cries from myriad throats)." (Lovecraft [29],[1925]; 546)

Certainly, Lovecraft attempts to induce a sensory experience in the reader, with sight as the most used sense, followed by hearing and smell; touch seldom occurs, and taste is completely absent ([30]; 186). As I have already mentioned, the emotion of fear that Lovecraft intends to induce is not of a physical kind, although he employs the senses to provoke an unpleasant sensory experience, which is almost always characterized as indescribable. This indescribability is precisely what evokes a fear beyond physical terror.

Regarding sight, mathematics, and particularly geometry is used "to help build the mood for which he was striving" ([31]; 10), describing abnormal structures and dimensions to create a frightening ambience. That is what Ingwersen ([32]; 57) calls "the horror of geometry." References to uncanny mathematics, beyond our ability to comprehend, contribute to enhancing the cosmic insignificance inherent in Lovecraft's philosophy [33].

"Their arrangement was odd, and seemed to follow the symmetries of some cosmic geometry unknown to earth or the solar system." (Lovecraft [34],[1929]; 66)

"He had said that the geometry of the dream-place he saw was abnormal, non-Euclidean, and loathsomely redolent of spheres and dimensions apart from ours. [...] and twisted menace and suspense lurked leeringly in those crazily elusive angles of carven rock where a second glance shewed concavity after the first shewed convexity." (Lovecraft [35],[1928]; 51)

"Everything was shadowy pantomime, as if seen at a vast distance through some intervening haze—although on the other hand the newcomer and all subsequent comers loomed large and close, as if both near and distant, according to some abnormal geometry." (Lovecraft [36],[1939]; 534)

"But rather along the alien curves and spirals of some ethereal vortex which obeyed laws unknown to the physics and mathematics of any conceivable cosmos." (Lovecraft [37],[1933]; 133)

Art is another element that appears recurrently with the intention of generating the emotion of fear. In Lovecraft's work, ethnic objects (statuettes, idols, or fetishes) "are usually indicators of things far worse than mere murder" (Hefner ; 667–668). In fact, the way Lovecraft induces the emotion of fear is similar to that employed with mathematics and geometry: detailed descriptions with extensive use of adjectives to show us all the fear and evilness that art objects exude:

"No recognized school of sculpture had animated this terrible object, yet centuries and even thousands of years seemed recorded in its dim and greenish surface of unplaceable stone. The figure, [...] represented a monster of vaguely anthropoid outline, but with an octopus-like head whose face was a mass of feelers, a scaly, rubbery-looking body, prodigious claws on hind and fore feet, and long, narrow wings behind. This thing, which seemed instinct with a fearsome and unnatural malignancy, was of a somewhat bloated corpulence, and squatted evilly on a rectangular block or pedestal covered with undecipherable characters." (Lovecraft [35][1928]; 41)

"Sculptured images of these shoggoths filled Danforth and me with horror and loathing. They were normally shapeless entities composed of a viscous jelly which looked like an agglutination of bubbles; and each averaged about fifteen feet in diameter when a sphere. They had, however, a constantly shifting shape and volume; throwing out temporary developments or forming apparent organs of sight, hearing, and speech in imitation of their masters, either spontaneously or according to suggestion." (Lovecraft [39][1936]; 278)

As far as the sense of hearing is concerned, Lovecraft's work contains numerous allusions to sounds, always of an unpleasant nature, which represents the total terror of the unknown [40]. The most characteristic element of Lovecraft's work is perhaps his teratonyms ([41]; 127), based on "sounds-patterns that lie outside English phonetics or run contrary to the phonotactics of the language to result in anti-aesthetic constructions." There exists "a preponderance of low and back vowels, that speakers perceive as dark, heavy, ugly, hostile, etc." ([42]; 35. Cited in Ref. [41]), to which the author adds consonants "that are perceived as harsh and grating" [41]; 133). The result is an awkward combination of sounds to pronounce and equally unpleasant to read, an effect that is reinforced by other typographical peculiarities and atypical hyphenation schemes [41]; 133). Some examples are found in the invocations to Shub-Niggurath and Cthulhu, ("Iä! Iä! Shub-Niggurath!", "Iä! Iä! Cthulhu fthagn!" or "Ph'nglui mglw'nafh Cthulhu R'lyeh wgah-nagl fhtagn"), which appear in several of his tales, in terms whose meaning is not clear (n'gha'ghaa, naf'lthagn or y'hah), and in the names of different creatures (S'gg'ha, S'ngac, Pth'thya-l'yi) or places (Ooth-nargai, Y'ha-nthlei). These teratonyms transmit "a sense of the huge, dark, gelatinous monstrosity that characterizes the entities described in his works" ([42]; 36. Cited in Ref. [41]). Additionally, Lovecraft's names are undecipherable or unpronounceable, reinforcing, again, human limitations.

Finally, the sense of smell seems to have deserved less attention than the previous ones. There exists a link between evil forces and smell, or an association of smell with fear and disgust, as Lovecraft frequently "describes inorganic matter (e.g., stone) in a way that suggests organic corruption or similar processes. Generally, these are accompanied by nauseating smells" [30]; 183). Perhaps the clearest example of the connection between evil forces and smell is that provided by one of the fragments of the *Necronomicon* included in *The Dunwich Horror*:

"By Their smell can men sometimes know Them near, but of Their semblance can no man know, saving only in the features of those They have begotten on mankind; and of those are there many sorts, differing in likeness from man's truest eidolon to that shape without sight or substance which is Them." (Lovecraft [34]; [1929]; 64)

However, beyond the sensorial experience, there is another key element in the process of inducing fear: time. As the author himself points out, "[t]he reason why time plays a great part in so many of my tales is that this element looms up in my mind as the most profoundly dramatic and grimly terrible thing in the universe. Conflict with time seems to me the most potent and fruitful theme in all human expression." (Lovecraft [83] [1937]; 1)

As a result of this insight, the time of narration overlaps with time on a universal scale, that is, a time before which "the entirety of human history does not measure, [...] a scale which the human mind cannot conceive" [43]; 837) and that, in the end, produces a shrinking effect on human nature that only intensifies the awareness of fear.

2.2. The emotions of trust and sadness

Trust and sadness also play a significant role in Lovecraft's work, as several scholars have pointed out. The quantitative analysis developed in section 4 reinforces this idea as well. In his correspondence with Clark Ashton Smith, Lovecraft mentions that "No weird story can truly produce terror unless it is devised with all the care and verisimilitude of an actual hoax" (H. P. Lovecraft to Clark Ashton Smith October 17, 1930. Cited in Joshi [44]; no page). Hence, he used several techniques "to achieve these hoaxes, including: realism and naturalness of characters, language and setting. Perhaps his most distinctive technique for capturing a sense of realism was to integrate the sciences within his fiction as elements of the setting and atmosphere" [45]; 30).

However, Henderson ([46]; 96) suggests another reason: in some of his writings, as in *The Case of Charles Dexter Ward*, his longest work of fiction, "by incorporating letters into the text, Lovecraft plays to his strengths, for they allow him to bypass interactions that would expose the flimsiness of his characters." In any case, the narrators in Lovecraft stories are often "taciturn academics passively observing the horrors that unfold" ([47]; 50), maybe because these types of profiles are usually perceived as objective and credible observers.¹

Finally, another key emotion in Lovecraft's narrative is sadness. The atmosphere of sadness is a fundamental element throughout the story, as the author's fiction focuses "on producing strong atmospheres of death, darkness and melancholy" ([48]; 6). However, the feeling of sadness is relevant not only by itself but also by its combined effect with fear. According to Plutchik's Wheel of Emotions, the blend of fear and sadness generates despair, an emotion partly based on the idea that there may be a beyond, but that it is inhabited by evil.

3. Method

The proposed method aims at classifying the stories into similar groups (clusters), using the emotions produced as a grouping criterion. The identification of these clusters enables, on the one hand, a definition of the emotional pattern in the tales, while, on the other, associates them with the three periods in which the author's literary production is usually classified: *the Macabre* (1905–1920), *the Dream Cycle* (1920–1927), and *the Cthulhu Mythos* (1925–1935).

All of Lovecraft's works analyzed in this paper have been obtained from *The H.P. Lovecraft Archive*, which includes a list of Lovecraft's works, revisions, minor works, and collaborations. I have focused specifically on his fiction texts, with two exceptions: in the first place, I have not taken into account the stories written in collaboration with other authors, as I believe they would distort the stylistic analysis; secondly, I have not considered those texts that are not extant or are too short to use for quantitative analysis (the shortest story included is *The Cats of Ulthar*, with 1346 words), because the methodology chosen for this work might lead to weak results when applied to texts that are too short. The final list of works analyzed is shown in [Appendix 1](#) and includes fifty-four tales, novelettes, and novels.

The quantitative analysis was conducted in three stages. First, I have analyzed each work in order to identify the relative percentage of emotions it conveys according to Plutchik's Wheel of Emotions. The Bag-of-Words (BoW) algorithm used in this paper determines the occurrence of words (tokens) within a document. Then, the algorithm associates each token to a certain emotion according to an equivalence dictionary (lexicon). For instance, in the lexicon used in this paper the word *monster* relates to the emotion of fear; *moribund* is associated with sadness, while *admiration* is associated with joy and trust. In this way, the greater or lesser weight of the different emotions throughout the document can be determined. Therefore, a text with numerous tokens associated with fear and few tokens related to Plutchik's other seven emotions will transmit fear with great intensity since its relative weight is high. I have conducted a sentiment analysis, structured in two steps, for each novel, according to the usual procedure in BoW algorithms. As a first step, the data have been processed, working on all the words that make up each work: transformation of capital letters to lowercase, elimination of punctuation marks, elimination of stop words or empty words (words that have no meaning in themselves, as they modify or accompany others, such as articles, pronouns or prepositions) and tokenization (a division of the text into smaller pieces or tokens, which in the case of this paper are words). Subsequently, the NRC lexicon of emotions ([20] [21]) links each token with the corresponding emotion. To put it simply, NRC is a high-quality lexicon that consists of a list of English words and their associations with Plutchik's eight basic emotions (anger, fear, anticipation, trust, surprise, sadness, joy, and disgust). It is, essentially, a dictionary of equivalences between words and the emotions created by them, and probably the most widely used one for this type of research (currently cited in about 2000 papers in all fields of knowledge). Mohammad and Turney use the Macquarie Thesaurus [49] as the source for unigrams and bigrams, and the annotations were manually done by crowdsourcing (defined by Ref. [21]; 437) as "the act of

¹ The reader interested in delving into the rhetorical devices used by Lovecraft to enhance the credibility of his stories can consult the recent work of Hernández Roura (2019).

breaking down work into many small independent units and distributing them to a large number of people, usually over the web”).

After this procedure was completed, the relative weight of each of the eight emotions defined by Plutchik’s model was calculated for each story. For example, in the tale “From Beyond,” the dominant emotions are fear (19.5%), trust (15.5%), and sadness (14.2%), followed by anger (13.6%), anticipation (10.7%) and disgust (9.8%). Joy and surprise present the lowest weights, 8.6%, and 8.2% respectively. As a result, in this tale, fear accounts for 19.5%, clearly being the emotion with the highest weight, followed at a considerable distance by trust and sadness. Therefore, for each tale, I have obtained a vector of emotions with eight elements, each one being the weight that each emotion has in it. The analysis of the fifty-four stories included results in a matrix with fifty-four rows and eight columns: each row represents the emotional pattern of a particular tale, and each column renders that emotions’ relative weight in the different stories. We will call it the Emotion Matrix (EM).

Once this process was finished, an agglomerative hierarchical cluster analysis was carried out to identify groups of works, that is, clusters of works that share similar characteristics from the point of view of the emotions they transmit. Cluster analysis is the method of classifying a set of objects (tales in this work) in such a way that the elements included in one group (or cluster) are more similar to each other than to those contained in other clusters. In this paper, the criterion for determining how similar the different stories are is their emotional pattern, that is, the relative weight of Plutchik’s eight basic emotions (anger, anticipation, disgust, fear, joy, sadness, surprise, and trust). For instance, two tales in which the dominant emotions are fear, sadness, and trust will be emotionally akin to each other, and therefore would be expected to be grouped in the same cluster; other tales in which the dominant emotions are joy and surprise will probably fall outside this cluster. This approach allows us to classify the different tales into homogeneous groups, from the perspective of the emotional pattern.

To perform this analysis, the Euclidean distance and Ward’s linkage method were used to identify compact clusters. Euclidean distance is the most common measurement and can be calculated as the square root of the sum of the square differences. Conceptually, it represents the distance between two points joined by a straight line. That is, if we consider the points X and Y and join them by a straight line, the length of that line is the Euclidean distance. In this work, each point is a tale, its coordinates are the relative weights in each of the eight emotions, and the Euclidean distance is the length of the straight line that joins one tale to another and is given by how close or far apart their coordinates are. With respect to the linkage method, I have chosen Ward’s one. The procedure operates as follows. At the initial step, each cluster is a single point (in this case, a tale). Then, the algorithm runs tests to see which clusters it should merge. The criterion is to select the pair of clusters that lead to a minimum increase in total within-cluster variance after merging. The within-cluster variance is determined as the weighted squared distance between cluster centers. Then, these clusters are merged. The process is repeated iteratively to classify all the points, and the result is represented graphically in a dendrogram or heatmap, which will be described later. The reason for choosing Ward’s method is that it tends to generate compact groups, which is an advantage for classification purposes. Once the different groups were identified, a boxplot of each of the emotions was constructed to facilitate the visual interpretation (a method for graphically depicting groups of numerical data through their quartiles).

All quantitative analyses were conducted in R [50], a programming environment oriented to statistical analysis, using several packages to elaborate the code: *reshape2* [51], *NbClust* [52], *tidytext* [53], *ggplot2* ([85]), *fmsb* [54], *scales* ([86]), *tidyr* [55], *stringr* [88], *gplots* [56], *dplyr* [89], and *readtext* [90] 57.

4. Results

First, to exemplify the analysis procedure, I will study two specific works: *The Whisperer in Darkness* (1930), and *At the Mountains of Madness* (1931). The relevance of these tales lies in, as Arroyo-Barrigüete concludes, these stories present the most connections with other narratives, which makes them a central piece in the process of creation of Cthulhu Mythos.

The emotional pattern of both stories (Fig. 1) allows us to draw some conclusions. Firstly, the emotions of fear, sadness, and trust have a weight close to 50%, which confirms that, at least in these stories, they are very relevant emotions. This result is consistent with the outcome of previous research mentioned in the literature review section: the emotions of fear, sadness, and trust play a key role in

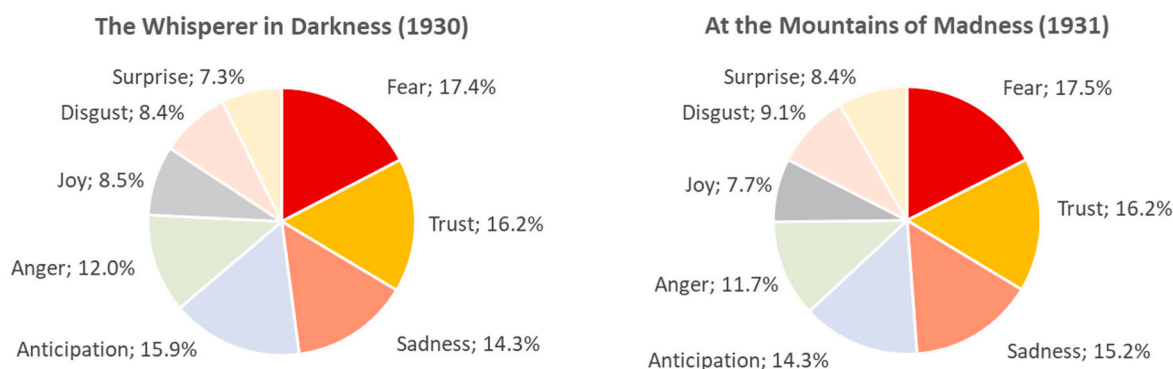


Fig. 1. Emotional pattern in *The Whisperer in Darkness* (1930), and *At the Mountains of Madness* (1931).

Lovecraft's work. However, the most interesting result is the remarkable similarity of the emotional pattern in both tales. As I mention below, the two are grouped in the same cluster (see appendix 1) for this reason. Other stories, sharing the general scheme (fear-sadness-trust) present quite different weights. For example, in the case of *The Very Old Folk* (1927), fear has a significantly higher weight (22.2%), while trust drops to 13.4%.

The study of the entire corpus led me to similar conclusions: the quantitative analysis reveals that fear is the dominant primary emotion, followed by sadness and trust, and the combination in dyads (composed of two emotions) produces emotions of despair and submission. I have selected a few excerpts from different tales that are particularly clear, simply to illustrate this point. Later in this section (see Fig. 2) the analysis of the whole corpus verifies the quantitative importance of fear in all the stories. The first fragment corresponds to the conclusion of *The Call of Cthulhu*, a very illustrative example of how the combination of fear and sadness generates despair:

"Death would be a boon if only it could blot out the memories. [...]"

I have looked upon all that the universe has to hold of horror, and even the skies of spring and the flowers of summer must ever afterward be poison to me. But I do not think my life will be long. As my uncle went, as poor Johansen went, so I shall go. I know too much, and the cult still lives.

Cthulhu still lives, too, I suppose [...]. Loathsomeness waits and dreams in the deep, and decay spreads over the tottering cities of men [...]" (Lovecraft [35]; 53–54)

On the one hand, the quantitative emotional analysis of this tale's ending fragment shows that the two prevailing emotions are fear (28%) and sadness (22%), and the reading of the narrative conveys considerable despair. On the other, the focus on the opening fragment of this tale points out to fear (21%) and trust (21%) as dominant emotions. Thus, the following extract invites readers to accept the existence of a superior power ("the cosmic cycle"), recognizing the insignificance of the "world and human race," which are merely "transient incidents," a truth so terrifying that the narrator confesses that "chills me when I think of it and maddens me when I dream of it."

"Theosophists have guessed at the awesome grandeur of the cosmic cycle wherein our world and human race form transient incidents [...]. But it is not from them that there came the single glimpse of forbidden aeons which chills me when I think of it and maddens me when I dream of it. That glimpse, like all dread glimpses of truth, flashed out from an accidental piecing together of separated things—in this case an old newspaper item and the notes of a dead professor. I hope that no one else will accomplish this piecing out; certainly, if I live, I shall never knowingly supply a link in so hideous a chain [...]" (Lovecraft [35]; 36)

Words like "dread," "accidental" and "hideous" are responsible for conveying the emotion of fear, while "professor," "truth" and "hope" convey the emotion of trust. Lovecraft uses trust with a twofold aim. The first is to provide credibility to the tale. Given that the events described are truly bizarre, this patina of verisimilitude promotes a reader's willing suspension of disbelief. In this story the support comes from a chronology of information compiled from reliable sources, such as an erudite university professor. In fact, the credentials of the individual voices are documented throughout the narrative, and they highlight their reliability ([58]; 42–43). The second purpose is to invoke, in combination with the emotion of fear, a feeling of submission.

Once the EM is calculated for the fifty-four works using the BoW algorithm previously described, the cluster analysis has been represented in a heatmap (Fig. 2). The interpretation of the heatmap shown in Fig. 2 is as follows. First, the shades of grey indicate the relative weight of each of the eight emotions for each tale, with white corresponding to the highest levels and black to the lowest ones.

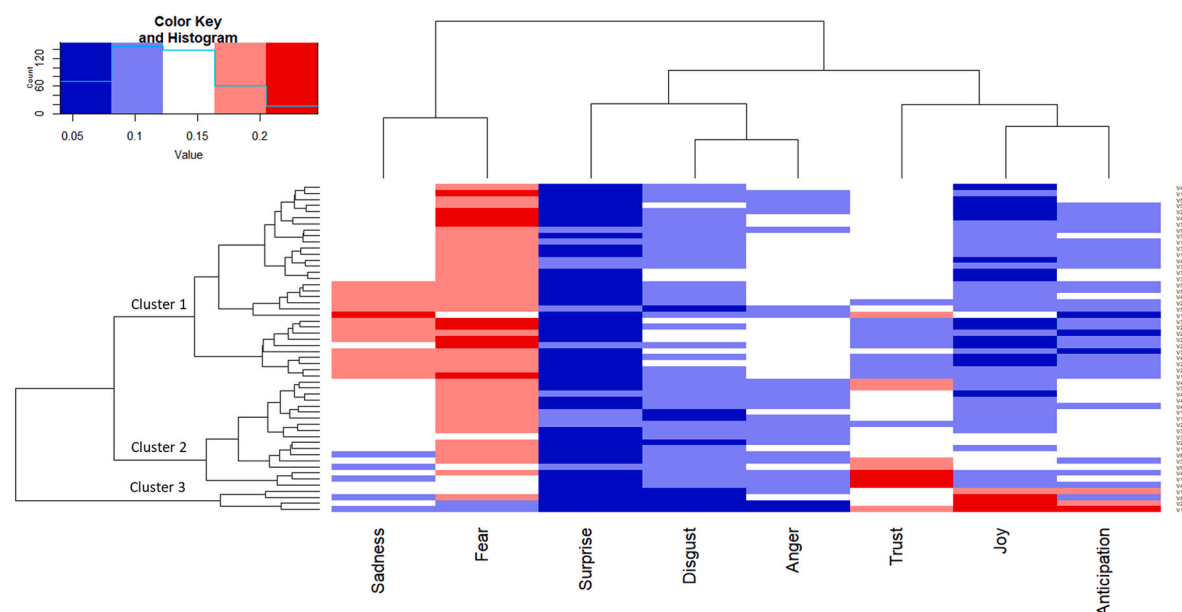


Fig. 2. Cluster analysis of the fifty-four selected works (heatmap).

The vertical interpretation shows that, while the remaining six emotions are associated more heterogeneously, fear and sadness constitute a compact group (fusion of its vertical lines at a relatively small height, and that joins with the rest of the emotions much later). In other words, as expected, fear and sadness are intimately linked in Lovecraft's works, and this association, as I have mentioned before, and according to Plutchik's theory, produces the emotion of despair so characteristic of his tales.

The horizontal interpretation, far more interesting for this article's perspective, makes clear how the tales are grouped. The horizontal lines on the left side of the figure display the distances between them, while the horizontal lines show when these works are grouped together in the same cluster. To determine the optimal number of clusters, twenty-four different indices have been compared, and the results indicate that seven of them recommend two clusters, ten recommend three clusters, and the remaining seven recommend between five and fifteen clusters. Thus, according to the majority rule, I choose three groups (see [Appendix 1](#)): Cluster 1 (thirty-two works), Cluster 2 (eighteen works), and Cluster 3 (four works). In fact, Cluster 3 is very different from the other two clusters, not only because it includes only four works but because according to the present analysis, it presents a pattern of emotions that is notably different from the rest of the works. Cluster 3 includes *The Doom That Came to Sarnath* (1919), *The White Ship* (1919), *Celephaïs* (1920) and *The Quest of Iranon* (1921). Due to the sample size, it is not possible to conduct a statistical analysis of this cluster, but a few interesting peculiarities can be observed, such as abnormally high levels of joy. These four works belong to the so-called Dream Cycle, and more specifically to the first years of that period. The fact that later works of this cycle, such as *Hypnos* (1922), *The Silver Key* (1926) or *The Dream-Quest of Unknown Kadath* (1926–27) are classified in other clusters, suggests two different conclusions. First, the distinction between the Dream Cycle and the *Cthulhu Mythos* is far from clear since both stages share many elements from the emotional point of view. Second, there is an indication that Lovecraft's works seem to evolve towards increasingly negative feelings, an insight I will discuss in detail later.

Regarding the suitability of the clusters obtained, I will consider as an example *The Call of Cthulhu* (coded as V36 in [Appendix 1](#)), Lovecraft's iconic work. According to the heatmap shown in [Fig. 2](#), the closest tales are *The Thing on the Doorstep* (V52) and *From Beyond* (V16), which form a compact sub-group since the horizontal lines merge relatively soon into one. [Fig. 3](#) shows the bar chart corresponding to the three stories above mentioned. The height of the bars represents the relative weight of the different emotions in each of them. This figure confirms that, from the point of view of emotions, they are almost identical.

All the tales belonging to Cluster 1 have been selected and the quartiles and maximum and minimum values for each of the eight emotions have been calculated to draw the corresponding boxplot. In [Fig. 4](#), the boxplot for Cluster 1 is shown, and the interpretation of the figure is as follows. For each of the eight emotions, we can find a vertical line, the ends of which indicates the minimum and maximum values. The box represents the first quartile (lower horizontal line), the median (inner horizontal line), and the third quartile (upper horizontal line). The points represent outliers, i.e., abnormally high or low percentages of that feeling. For instance, focusing on the emotion of fear, we observe that the first quartile is around 0.19, the median close to 0.2, and the third quartile somewhat below 0.21. This means that in 50% of the stories that make up Cluster 1, the emotion of fear has a relative weight of between 0.19 and 0.21 approximately. Another 25% have weights slightly below 0.19 and another 25% have values slightly above 0.21, within the limits marked by the vertical line.

As seen, in the thirty-two works that make up this cluster, the dominant emotions are, in the first place, fear, followed by sadness, and trust, although anger also ranks quite high. Surprise and joy are the two emotions with the least weight. Performing the same exercise for the eighteen works in Cluster 2, we observe a different pattern because although the three dominant feelings are the same, their order of importance is different, it being fear, trust, and sadness. Additionally, the relative weight of fear in Cluster 2 is lower than that in Cluster 1, and the level of joy is significantly higher.

The comparison of both patterns (the emotional pattern of Cluster 1 and that of Cluster 2) leads to some interesting conclusions.

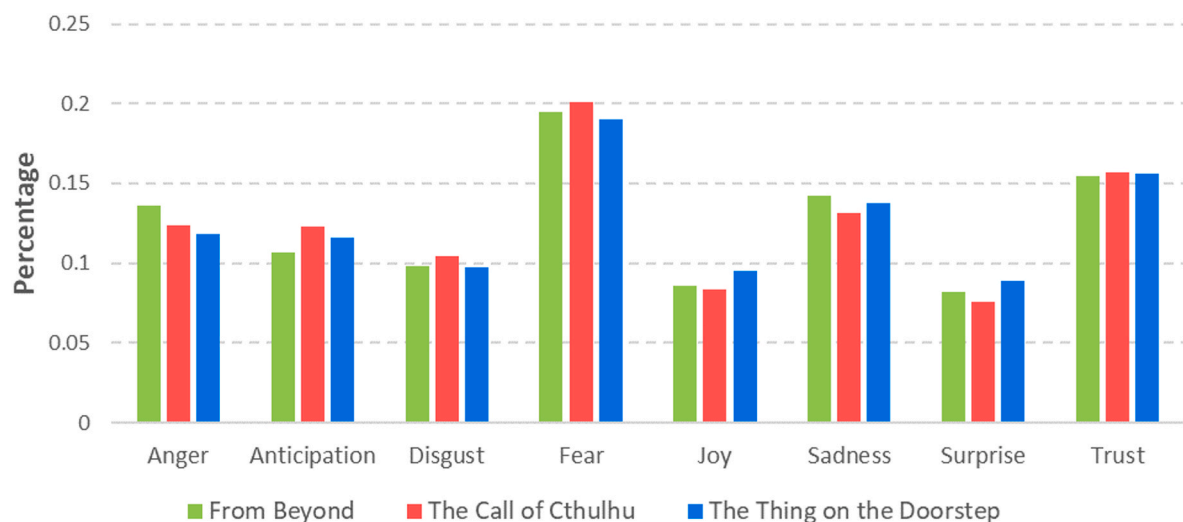


Fig. 3. Relative weight of each of the emotions in *The Call of Cthulhu* and the two closest stories according to the cluster analysis.

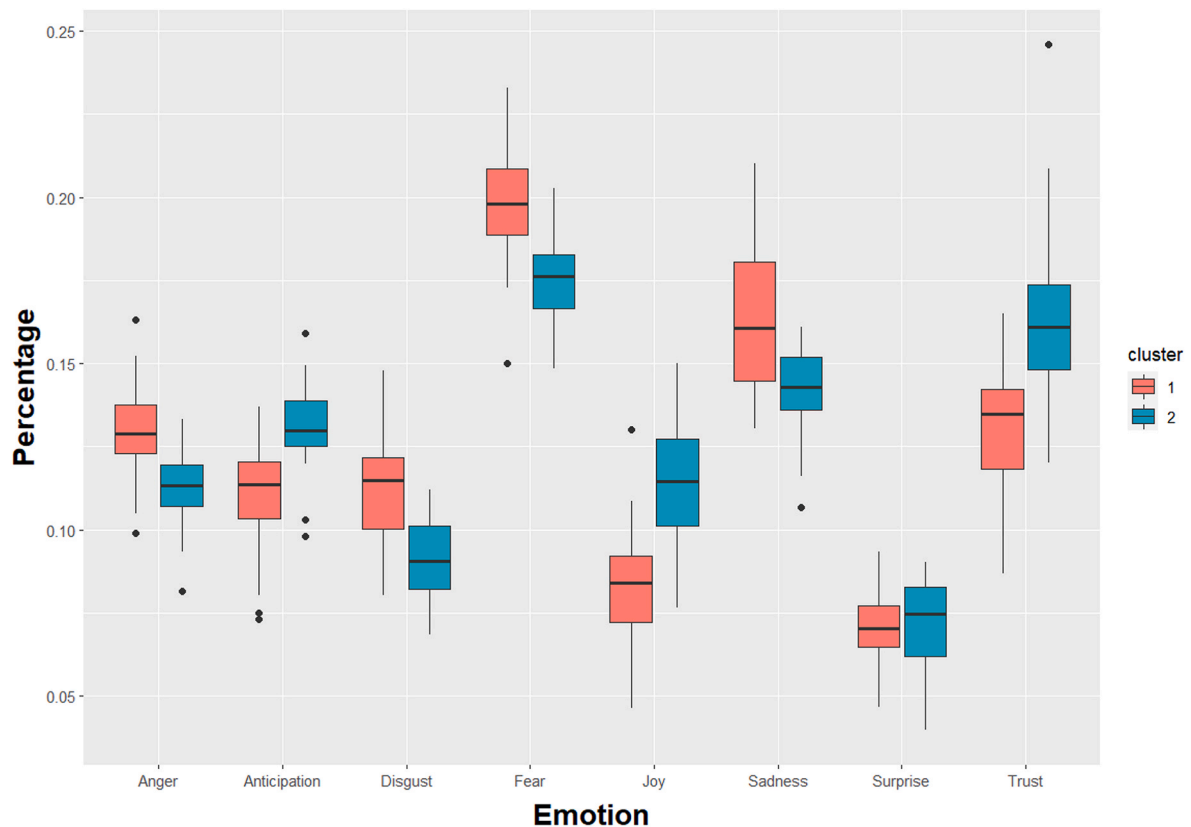


Fig. 4. Boxplot of emotions comparing Clusters 1 and 2.

First, it is noticeable that in Cluster 2, there is a better equilibrium, while in Cluster 1, there is a greater polarization, with strong differences in the relative weight of the emotions. In other words, the stories in Cluster 1 present a lower emotional balance, with one emotion, fear, outscoring the others. In fact, the median is close to 20%, which implies a disproportionately heavy weight of this emotion. It is almost double what it would be if all emotions were of equal importance. For example, *The Call of Cthulhu*, *The Thing on the Doorstep*, and *From Beyond*, which are part of this cluster, present fear levels of 20.1%, 19.0%, and 19.5% respectively. The emotional imbalance is clearly apparent when compared to the emotions of joy or surprise, which do not ever reach 10% in any of these stories. Even if we carry out the comparison with the second most important emotion in these tales, sadness, we observe that it is significantly below the levels of fear: 13.1%, 13.8%, and 14.2% respectively. Second, we can see how the works in Cluster 1 are darker, with a clear turn towards negative emotions: fear, sadness, anger, and disgust increase significantly when compared to Cluster 2, while joy reduces its relative weight considerably in Cluster 1. Third, although trust remains a key emotion in both clusters, it has a significantly lower weight in cluster 1, which is relevant when interpreting the combination of emotions: in Cluster 2, the combination of the two main emotions (fear and trust) generates the emotion of submission according to Plutchik's theory. However, in Cluster 1, the combination of the two fundamental emotions (fear and sadness) produces despair in Plutchik's model.

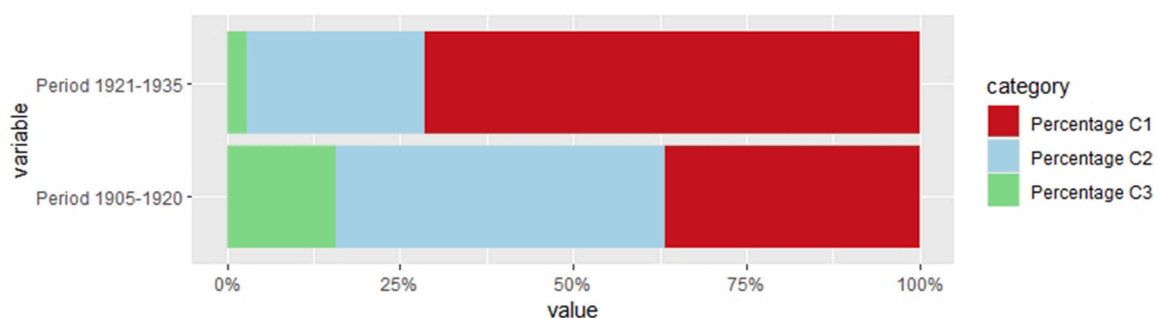


Fig. 5. Relative percentage of Lovecraft's works in each cluster differentiating between the two periods.

These differences lead to the question of whether there is any relationship between the clusters and the classification of Lovecraft's work into two distinct periods already mentioned in the introduction. To evaluate this point, all the stories were separated into two periods, 1905–1920 and 1921–1935. In both periods the percentage of tales belonging to each cluster was calculated. As shown in Fig. 5, in the period 1905–1920, only 37% of the works follow the pattern of Cluster 1, while 71% of the works written between 1921 and 1935 follow this pattern. In other words, there seems to be a considerable difference between the two periods. This finding has been confirmed by carrying out a unilateral contrast for the difference in proportions. Since the proportions are very different in both periods, it would seem that there is indeed a difference between them. However, to be certain, it is necessary to carry out a hypothesis test. This statistical technique allows us to confirm or refute with a certain level of confidence (95% in this case) whether or not the difference observed between the two percentages can be considered statistically significant. I have obtained a p-value of 0.015, below 0.05. Therefore, at a confidence level of 95%, Lovecraft's writing pattern with regard to emotion is different in both periods, with a statistically significant evolution toward greater use of the pattern corresponding to Cluster 1. That is, evolution is observed in Lovecraft's style toward works that are more polarized in terms of feelings, are darker, with a greater relative weight on the emotions of fear, sadness, anger, and disgust, and in which the key emotions are fear and sadness that become despair when combined.

5. Conclusions

Throughout this paper, fifty-four works of Lovecraft have been analyzed, which span almost all of his career as a fiction writer. Using computational linguistics techniques, various conclusions have been reached in relation to his work. In the first place, the research shows that three emotions dominate the corpus analyzed: fear, trust, and sadness. This result is consistent with the conclusions of Arroyo-Barrigüete, who analyzed four tales (*The Call of Cthulhu*, *The Mound*, *The Shunned House*, and *The Case of Charles Dexter Ward*) and concluded that in these stories fear, trust, and sadness are precisely the dominant emotions. The present research, precisely by extending the assessed corpus to fifty-four stories, confirms that this is a general pattern in Lovecraft's work, and not exclusive to the aforementioned stories. Fear is clearly the dominant emotion. This result, which I have quantitatively confirmed, is consistent with the findings of previous research about the importance of fear in Lovecraft's work, among others [16,32[91]]. Also of interest is the status of the emotion of trust. One might expect such emotion to play a very relevant role in Lovecraft's work, for as the author himself pointed out "[n]o weird story can truly produce terror unless it is devised with all the care and verisimilitude of an actual hoax" (cited above). Through quantitative techniques of analysis, I have been able to confirm that this statement of the author's intent does translate to his writing style. Finally, the emotion of sadness also plays a very relevant role, in line with Norman's findings ([48]; 6), who pointed out that Lovecraft's fiction focuses "on producing strong atmospheres of death, darkness and melancholy."

The second conclusion of this work refers to an existing intimate connection between fear and sadness, which has been confirmed by the quantitative analysis, while the other six emotions of Plutchik's Wheel of Emotions appear to be grouped somewhat more heterogeneously.

In the third place, I would like to assert that the vast majority of his works follow two patterns as far as emotions are concerned. The first group (Cluster 1) is characterized by a considerable polarization of emotions, with fear being the dominant one, followed by sadness and trust. These are very dark writings with an overwhelming preponderance of negative emotions. The second group of works (Cluster 2) presents a greater emotional balance. This cluster shows the same three leading emotions, but their order of importance is different: fear, trust, and sadness. These writings are less dark than those corresponding to Cluster 1, with a lower weight of negative emotions such as fear, sadness, anger, and disgust and higher relevance of the feeling of joy. In addition, there is a third group, made up of only four works that are notably different from the rest since the emotion of joy is prevailing. The existence of these plainly differentiated patterns is interesting because there seems to be a relationship between them and the evolution of Lovecraft's style, which leads to my last conclusion: it is not feasible to establish clear limits in Lovecraft's production. The three periods mentioned in

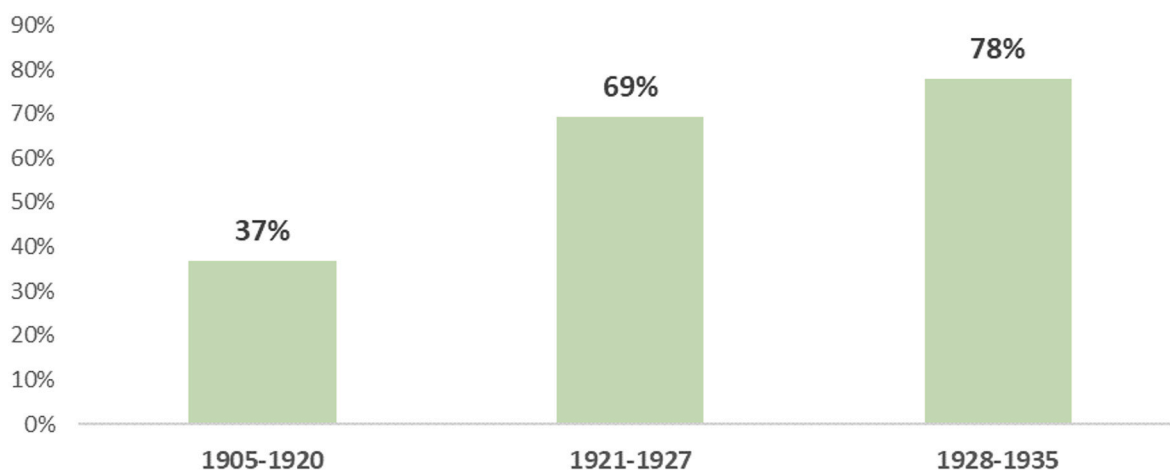


Fig. 6. Percentage of works belonging to Cluster 1 in Lovecraft's three periods.

the introduction present blurred boundaries, and therefore, it seems more appropriate to speak of progressive evolution. From a thematic point of view, it is difficult to categorize the stories by their dates of creation [5]. The results of my analysis suggest that, from an emotional analysis perspective, this difficulty is even greater. Analyzing the list of works and the cluster to which they belong (Appendix 1), I can confirm that it is not possible to establish a clear correspondence between the different periods and the cluster to which each work belongs, except perhaps between *The Macabre* (1905–1920) and the rest of his tales, as has been indicated. What can be observed (Fig. 6) is that the percentage of works belonging to Cluster 1, the darkest from an emotional point of view, increases over time, with a disruptive leap being observed from 1920 onwards.

The comparison of the periods 1905–1920 (*the Macabre*) and 1921–1935 (*the Dream Cycle* and *the Cthulhu Mythos*), shows that in his first period, only 37% of the works follow the pattern of Cluster 1, while in the second one 71% of the stories does. So, there is an evolution in Lovecraft's narrative style, which tends more and more towards darker stories. Tales in the period 1921–1935 are dominated by negative emotions, in which despair, as the result of combining fear and sadness, plays a predominant role. So, it is worth asking to what extent there is a relationship between this transition and the author's own biography. Maybe, this progression is consistent with his state of mind, for as the correspondence of his later years reveals, Lovecraft showed growing hopelessness: "every aptitude which I wish I had, I lack. Everything which I wish I could formulate & express, I have failed to formulate & express. Everything which I value, I have either lost or am likely to lose" (excerpt from a letter sent to Helen Sully [59]). It is true, though, that other contemporary writers of Lovecraft also adopted a darker style just after the end of the First World War. My paper shows that there has been a transitional moment in Lovecraft's writing, but the analysis I have performed cannot determine its cause. I believe the application of comparative/contrastive corpora techniques will shed light on the origin of Lovecraft's main emotions, fear, and sadness, opening thus a door to a better understanding of the despair that inhabited his fiction.

Declarations

Author contribution statement

Jose Luis Arroyo-Barrigüete: Conceived and designed the experiments; Performed the experiments; Analyzed and interpreted the data; Contributed reagents, materials, analysis tools or data; Wrote the paper.

Funding statement

This research did not receive any specific.

Appendix 1. list of Lovecraft's fiction used in this paper

The dates for each of the tales have been obtained from The H. P. Lovecraft Archive [57] and refer to the time of their creation, which is in turn based primarily on [61]. Some changes based on Joshi's ongoing research have been included.

Code	Title	Year	Cluster
V1	The Beast in the Cave	1905	1
V2	The Alchemist	1908	1
V3	Dagon	1917	2
V4	A Reminiscence of Dr. Samuel Johnson	1917	2
V5	The Tomb	1917	1
V6	Polaris	1918	2
V7	Beyond the Wall of Sleep	1919	2
V8	The Doom That Came to Sarnath	1919	3
V9	Old Bugs	1919	2
V10	The Statement of Randolph Carter	1919	1
V11	The Transition of Juan Romero	1919	2
V12	The White Ship	1919	3
V13	The Cats of Ulthar	1920	1
V14	Celephaïs	1920	3
V15	Facts Concerning the Late Arthur Jermyn and His Family	1920	2
V16	From Beyond	1920	1
V17	The Picture in the House	1920	1
V18	The Temple	1920	2
V19	The Street	1920?	2
V20	The Moon-Bog	1921	2
V21	The Music of Erich Zann	1921	1
V22	The Nameless City	1921	1
V23	The Outsider	1921	1
V24	The Quest of Iranon	1921	3
V25	Herbert West—Reanimator	1921–22	1

(continued on next page)

(continued)

Code	Title	Year	Cluster
V26	The Hound	1922	1
V27	Hypnos	1922	2
V28	The Lurking Fear	1922	1
V29	The Festival	1923	1
V30	The Rats in the Walls	1923	1
V31	The Unnamable	1923	1
V32	The Shunned House	1924	1
V33	He	1925	2
V34	The Horror at Red Hook	1925	1
V35	In the Vault	1925	1
V36	The Call of Cthulhu	1926	1
V37	Cool Air	1926	1
V38	Pickman's Model	1926	1
V39	The Silver Key	1926	2
V40	The Strange High House in the Mist	1926	2
V41	The Descendant	1926?	2
V42	The Dream-Quest of Unknown Kadath	1926–27	1
V43	The Case of Charles Dexter Ward	1927	2
V44	The Colour out of Space	1927	1
V45	The Very Old Folk	1927	1
V46	The Dunwich Horror	1928	1
V47	The Whisperer in Darkness	1930	2
V48	At the Mountains of Madness	1931	2
V49	The Shadow over Innsmouth	1931	1
V50	The Dreams in the Witch House	1933	1
V51	The Evil Clergyman	1933	1
V52	The Thing on the Doorstep	1933	1
V53	The Shadow out of Time	1934–35	1
V54	The Haunter of the Dark	1935	1

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